

A Data-Driven Framework for County-Level Obesity Risk Identification, Forecasting, and Public Health Resource Prioritization in the United States

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Abstract - Obesity prevalence in the United States exhibits significant geographic and socioeconomic heterogeneity, making uniform public health interventions inefficient. This study proposes a data-driven framework to identify county-level obesity risk, analyze structural patterns, forecast future trends, and prioritize public health resources using publicly available health, socioeconomic, and demographic data. County-level datasets spanning 2010 to 2023 were integrated to construct two complementary analytical views. A cross-sectional snapshot centered on 2019 was used for exploratory analysis, spatial statistics, clustering, supervised learning, and decision modeling, while a longitudinal panel covering approximately fourteen years was used to derive temporal trends, early warning indicators, and long-term forecasts. Spatial analysis revealed statistically significant regional clustering of obesity prevalence. Predictive models explained a substantial portion of county-level obesity variation, and forecasting models were used to project trends through 2030. The results demonstrate a scalable decision-support framework for targeted, cost-effective public health intervention planning.

Keywords - Obesity Analytics, Data Mining, Spatial Analysis, Unsupervised Learning, Supervised Learning, Forecasting, Public Health Decision Support

I. INTRODUCTION

Obesity has emerged as one of the most significant public health challenges in the United States, contributing to increased prevalence of cardiovascular disease, diabetes, and reduced quality of life. In addition to its health consequences, obesity imposes substantial economic costs through increased healthcare expenditures and productivity losses. Despite national-level awareness, obesity prevalence varies widely across counties, indicating that localized socioeconomic, demographic, and behavioral factors play a critical role in shaping health outcomes. As a result, effective intervention strategies require county-level analysis rather than uniform statewide or national approaches.

Current obesity surveillance efforts often rely on descriptive statistics, statewide averages, or isolated indicators such as physical inactivity or income. While these approaches provide useful summaries, they fail to capture complex interactions between socioeconomic conditions, demographic characteristics, health behaviors, and geographic context. Moreover, many studies treat counties as independent units, ignoring spatial dependence and temporal trends that influence obesity outcomes over time. Consequently, decision makers lack integrated analytical tools that can both explain observed disparities and support proactive, targeted intervention planning.

To address these limitations, this work proposes an end-to-end data mining framework for county-level obesity analysis that integrates exploratory data analysis, spatial statistics, machine learning, and forecasting techniques. By combining multiple analytical perspectives, the proposed framework identifies structural obesity risk patterns, estimates future trends, and synthesizes results into an interpretable decision-support model designed for public health planning.

The primary contributions of this paper are as follows:

1. Integration of multi-source county-level datasets capturing health, socioeconomic, and demographic factors across the United States;
2. Identification of statistically significant spatial clustering in obesity prevalence using spatial autocorrelation methods;
3. Discovery of distinct county risk profiles through unsupervised learning techniques;
4. Prediction of obesity prevalence using supervised regression and time-based forecasting models; and
5. Development of a multi-criteria decision model to prioritize counties for targeted public health intervention.

II. DATA SOURCES, DATASET CONSTRUCTION, AND PREPROCESSING

A. Data Sources

This study integrates multiple publicly available datasets to construct a comprehensive county-level obesity analytics dataset for the United States. Health outcome variables, including obesity prevalence, physical inactivity rates, were obtained, socioeconomic indicators such as median household income, poverty rate, unemployment rate were sourced from the Rural Health Information Hub which is supported by the Health Resources and Services Administration (HRSA) of the U.S. Department of Health and Human Services (HHS). Additional contextual variables, including demographics population, were obtained from the census.gov

All datasets are aggregated at the county level and span the period from 2010 to 2023. County Federal Information Processing Standards codes were used as unique identifiers to ensure consistent alignment across data sources. The resulting data capture substantial geographic, socioeconomic, and behavioral heterogeneity across

approximately three thousand one hundred United States counties.

B. Dataset Construction

From the integrated data sources, two complementary analytical representations were constructed to support different stages of the analysis. First, a cross-sectional county-level snapshot centered on 2019 was created to serve as a stable baseline for exploratory analysis, spatial modeling, clustering, supervised learning, and multi-criteria decision modeling. The year 2019 was selected to avoid disruptions associated with pandemic-era reporting while providing a representative pre-pandemic baseline.

Second, a longitudinal panel dataset spanning approximately fourteen years was constructed to support temporal analysis. This panel representation enables the computation of trend-based features, including slope and rate-of-change indicators, as well as the development of early warning metrics and long-term forecasting models. Together, these two representations allow the framework to capture both structural and temporal dimensions of obesity risk.

C. Data Integration and Preprocessing

Because the datasets originated from multiple sources with differing formats, temporal coverage, and reporting standards, extensive preprocessing was required prior to analysis. County Federal Information Processing Standards codes were used to harmonize datasets, and variables were standardized to ensure consistent measurement scales across sources. Continuous variables were normalized where appropriate to facilitate comparability across modeling techniques.

Missing values were handled using median imputation for continuous variables, which provides robustness to skewed distributions commonly observed in socioeconomic data. Records with irreconcilable geographic mismatches were excluded to preserve data integrity. The resulting cleaned and integrated datasets form a reproducible analytical foundation suitable for exploratory analysis, spatial modeling, machine learning, forecasting, and decision-support applications.

III. EXPLORATORY DATA ANALYSIS AND SPATIAL ANALYSIS

A. Cross-Sectional Exploratory Data Analysis

Exploratory data analysis was first conducted using a cross-sectional county-level snapshot centered on 2019 to examine structural patterns in obesity prevalence and associated socioeconomic, demographic, and behavioral factors. The 2019 baseline was selected to provide a stable, pre-pandemic reference year for comparative analysis across counties.

Univariate analysis revealed substantial heterogeneity in obesity prevalence across U.S. counties, indicating that national or statewide averages obscure meaningful local variation. Similar dispersion was observed across key explanatory variables, including physical inactivity, median

household income, poverty rate, and educational attainment, suggesting that obesity risk is influenced by a combination of behavioral and socioeconomic conditions.

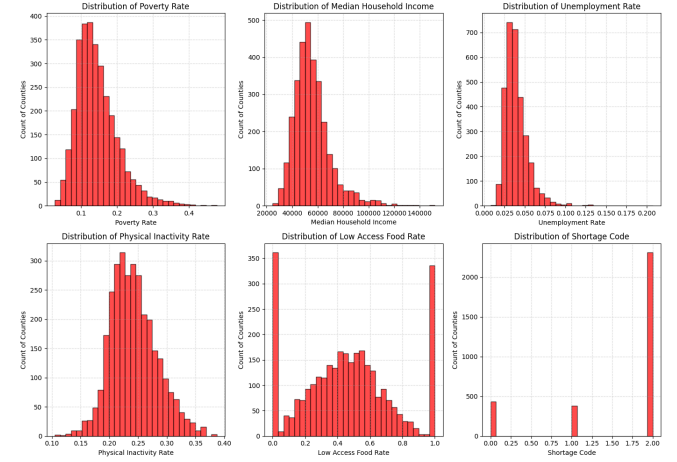


Fig. 1 Distribution of county-level features (2019), showing substantial variation across regions.

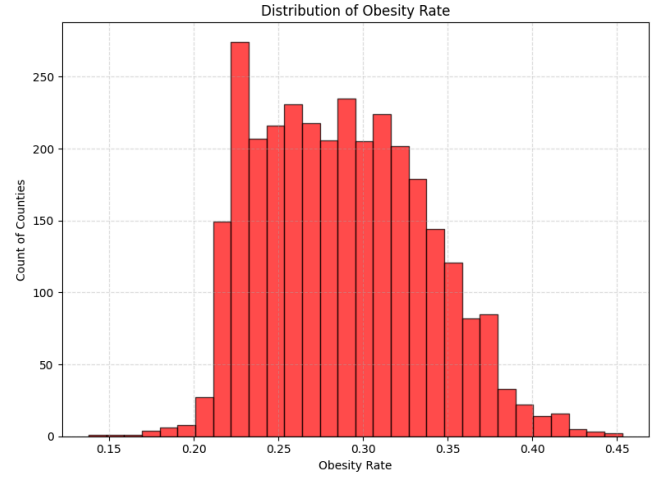


Fig. 2 Distribution of county-level obesity prevalence across the United States (2019), illustrating substantial heterogeneity that motivates localized analysis.

Bivariate analysis using correlation and scatter plots identified physical inactivity as the strongest positive correlate of obesity prevalence, while median household income exhibited a negative association. However, these relationships were not strictly linear, indicating the presence of interaction effects and threshold behavior. Mutual information analysis further confirmed that behavioral and socioeconomic features contributed more explanatory power than demographic variables alone.

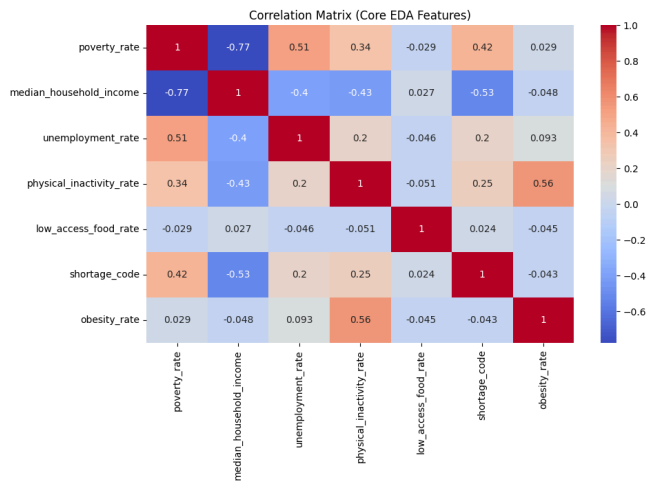


Fig. 3. Correlation matrix of core socioeconomic and behavioral variables with obesity prevalence, highlighting strong associations and interdependencies among predictors.

These findings highlight the limitations of single-factor explanations and motivate the application of unsupervised and supervised learning techniques in subsequent sections to capture complex, multivariate obesity risk patterns.

B. Temporal Exploratory Data Analysis

To complement the cross-sectional analysis, a temporal exploratory data analysis was conducted using a longitudinal county-level panel spanning approximately fourteen years. This analysis focuses on understanding how obesity prevalence and related indicators evolve over time rather than comparing counties at a single point.

Time-series visualization revealed that while obesity prevalence has increased nationally, counties exhibit markedly different growth trajectories. Some counties demonstrate steady, gradual increases, whereas others show accelerated growth rates or periods of rapid change. These divergent patterns suggest that static snapshots alone are insufficient to identify emerging high-risk regions.

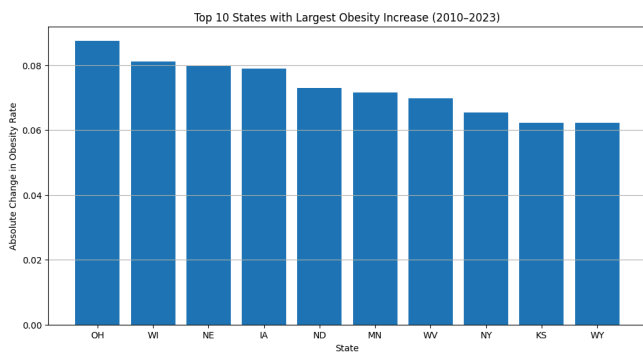


Fig. 4 States with the Largest Increase in Obesity Prevalence (2010–2023)

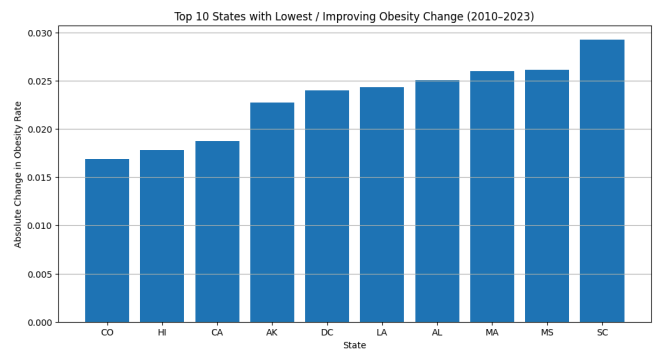


Fig. 5 represents top 10 states with improving obesity change(2010-2023)

To quantify temporal dynamics, slope-based trend features were computed for each county, capturing the direction and magnitude of obesity change over time. Counties with higher positive slopes indicate accelerating obesity risk, even when current prevalence levels remain moderate. This observation underscores the importance of incorporating trend-based indicators into risk assessment frameworks.

Temporal exploratory analysis thus provides early warning signals that are not detectable through cross-sectional analysis alone and directly motivates the development of forecasting and early warning models presented in subsequent sections.

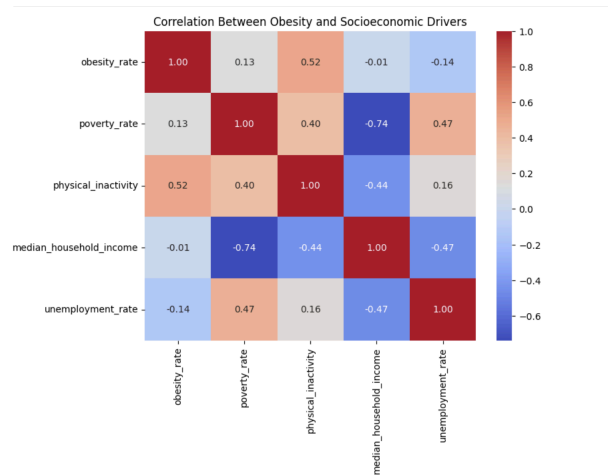


Fig. 6 Correlation Matrix Between Average Obesity Prevalence and Socioeconomic Indicators

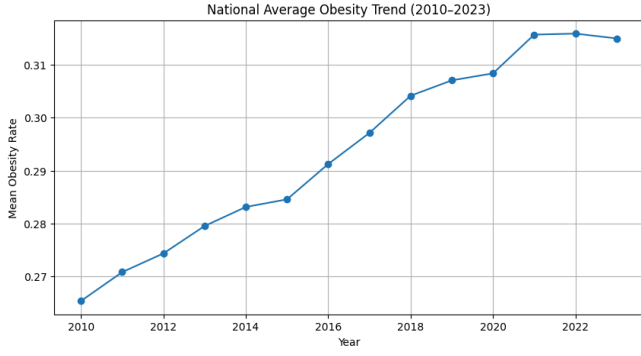


Fig. 7 National Average Obesity Trend Across U.S. Counties (2010–2023)

To establish a national temporal baseline, the average county-level obesity prevalence was computed for each year between 2010 and 2023. Figure 7 illustrates a clear and persistent upward trend in mean obesity prevalence across U.S. counties over the study period. National average obesity increased steadily from approximately 26.5% in 2010 to over 31% by 2021, with only minor stabilization observed in the most recent years.

This sustained increase motivates the need for temporal modeling beyond cross-sectional analysis, as it suggests that obesity risk is not static but evolving over time. While national averages provide a useful macro-level summary, they obscure substantial heterogeneity in county-level trajectories. Consequently, subsequent analyses focus on identifying counties with accelerating growth, divergent trend regimes, and early-warning signals that may not be apparent from aggregate national trends alone.

C. Spatial Distribution of Obesity Prevalence

Spatial visualization techniques were applied to the 2019 county-level obesity prevalence data to examine geographic patterns across the United States. Choropleth mapping revealed pronounced regional disparities, with elevated obesity rates concentrated in contiguous geographic regions rather than randomly distributed across counties. In particular, higher prevalence clusters were observed across portions of the southeastern and midwestern United States, while lower prevalence counties were more common in coastal and metropolitan regions.

These spatial patterns suggest that obesity prevalence is influenced by geographically shared socioeconomic, environmental, and behavioral factors. The observed regional concentration further indicates that treating counties as independent analytical units may overlook meaningful spatial dependence.

D. Spatial Autocorrelation and Hotspot Analysis

To formally assess spatial dependence, Global Moran’s I statistic was computed using county-level obesity prevalence for the 2019 baseline. The results indicated statistically significant positive spatial autocorrelation, confirming that counties with similar obesity rates tend to be

geographically clustered rather than randomly distributed. This finding validates the visual patterns observed in the spatial distribution analysis and underscores the importance of accounting for geographic context.

To identify localized clusters, Getis-Ord G_i^* hotspot analysis was applied. This method revealed statistically significant hotspots corresponding to regions of elevated obesity prevalence, as well as cold spots characterized by lower prevalence. The identified hotspots and cold spots align closely with broader regional trends, providing quantitative evidence of localized obesity risk concentrations.

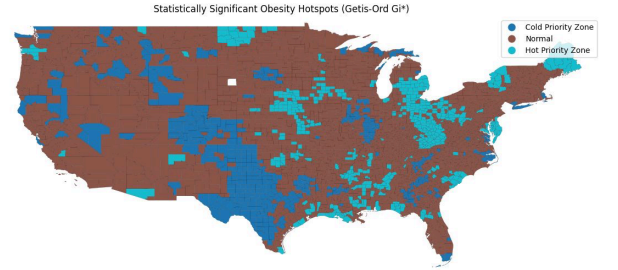


Fig. 8 Statistically significant obesity hotspots and cold spots identified using Getis-Ord G_i^* analysis for U.S. counties (2019). Hotspot regions indicate clusters of significantly elevated obesity prevalence, while cold spots represent significantly lower prevalence areas.

E. Implications for Subsequent Modeling

The presence of strong spatial clustering has important implications for the modeling stages that follow. First, it motivates the use of unsupervised learning techniques to identify groups of counties with similar obesity risk profiles beyond simple geographic proximity. Second, spatial heterogeneity reinforces the need for supervised models capable of capturing non-linear relationships between obesity prevalence and socioeconomic factors. Finally, the identification of geographic hotspots supports targeted public health intervention planning rather than uniform policy deployment.

IV. METHODOLOGY

A. Unsupervised Learning

Unsupervised learning was applied to the 2019 cross-sectional county-level dataset to identify latent obesity risk profiles without relying on predefined outcome labels. The goal of this analysis is to discover structurally similar groups of counties based on socioeconomic, behavioral, demographic, and health-related characteristics, thereby enabling data-driven segmentation for targeted public health analysis.

Prior to clustering, all continuous features were standardized to ensure comparability across variables with different scales. To reduce dimensionality while preserving non-linear structure in the data, Uniform Manifold

Approximation and Projection (UMAP) was applied to project high-dimensional county feature vectors into a lower-dimensional embedding, preserving non-linear structure while reducing noise and improving cluster separability. Clustering was then performed on the reduced feature space using a comprehensive model selection pipeline that evaluated multiple unsupervised algorithms and parameter configurations.

Specifically, K-means, agglomerative hierarchical clustering, Gaussian mixture models, spectral clustering, DBSCAN, and HDBSCAN were systematically explored across a range of hyperparameters. Model performance was assessed using internal validation metrics, including silhouette score, Davies–Bouldin index, and Calinski–Harabasz criterion, enabling comparison of cluster compactness, separation, and stability across methods. Based on the metrics, a two-cluster solution ($K = 2$) provided the best balance between cluster compactness and separation.

The resulting clusters reveal distinct county-level obesity risk profiles characterized by differing combinations of obesity prevalence, physical inactivity, income level, and poverty rate. One cluster exhibits consistently higher obesity prevalence coupled with adverse socioeconomic and behavioral indicators, while the other reflects relatively lower-risk profiles with more favorable conditions. Importantly, cluster membership does not strictly align with geographic proximity, indicating that obesity risk is shaped by shared structural conditions beyond spatial location alone.

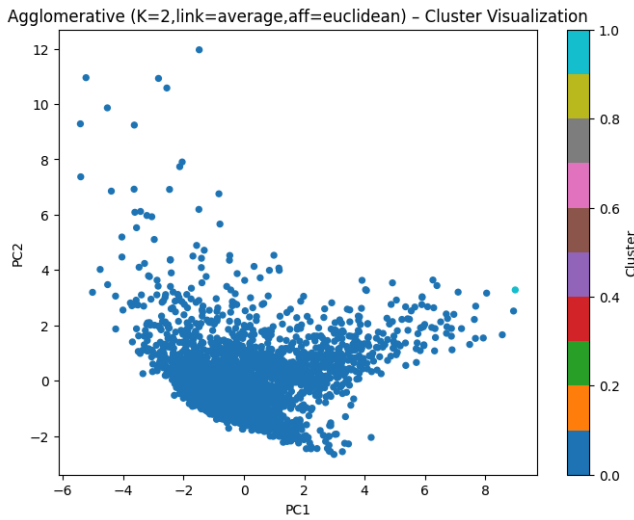


Fig. 9 Visualization of county-level obesity risk clusters in reduced feature space for all features using agglomerative clustering on the 2019 baseline dataset.

These unsupervised findings provide a continuous cluster, enabling for subsequent supervised learning and decision modeling stages by summarizing complex, multivariate patterns into a small number of meaningful county risk groups.

B. Supervised Learning

Supervised learning models were applied to the 2019 cross-sectional dataset to estimate county-level obesity prevalence and to quantify the relative importance of socioeconomic, behavioral, and demographic factors. Unlike unsupervised learning, which identifies latent structure, supervised models enable direct assessment of predictive performance and explanatory drivers of obesity risk.

Obesity prevalence was modeled as a continuous outcome variable using a suite of regression-based supervised learning algorithms. The modeling pipeline included both linear and non-linear approaches, specifically ordinary least squares linear regression, ridge regression, and tree-based ensemble models, including Random Forest Regression and Gradient Boosting Regression. This combination allowed evaluation of both interpretable linear relationships and more complex non-linear interactions among predictors.

Model performance was evaluated using cross-validation to ensure robustness and to mitigate overfitting. Among the models tested, the tree-based ensemble model (Gradient Boosting Regressor) achieved the strongest predictive performance, explaining approximately forty percent of the variance in county-level obesity prevalence on held-out data. This performance exceeded that of linear and regularized regression models, indicating the presence of meaningful non-linear relationships between obesity prevalence and its associated socioeconomic and behavioral drivers.

Overall, these results suggest that while linear socioeconomic gradients explain part of county-level obesity variation, non-linear interactions captured by ensemble models provide additional explanatory power. The supervised learning stage therefore establishes both a quantitative predictive baseline and an interpretable foundation for subsequent feature importance analysis, interaction assessment, and policy-oriented decision modeling.

Residual Map

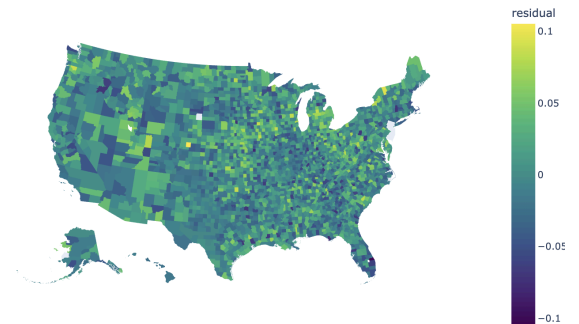


Fig. 10 Spatial distribution of supervised model residuals for county-level obesity prevalence (2019). Positive values indicate under-prediction, while negative values indicate over-prediction.

Feature importance analysis identified physical inactivity as the strongest predictor of obesity prevalence, followed by socioeconomic indicators such as median household income and poverty rate. These findings are consistent with patterns observed during exploratory data analysis, reinforcing the role of behavioral and economic conditions as primary drivers of county-level obesity risk.

By jointly evaluating predictive accuracy and feature importance, supervised learning provides actionable insight into modifiable risk factors and establishes a quantitative foundation for subsequent multi-criteria decision modeling and policy-oriented analysis.

C. Multi-Criteria Decision Modeling and Policy Decision Support

While supervised learning models provide predictive accuracy and feature importance, they do not directly translate into actionable prioritization for public health planning. Decision makers must allocate limited resources across regions based on multiple competing risk indicators rather than a single predicted outcome. To address this gap, a multi-criteria decision modeling framework was developed to synthesize results from prior analyses into an interpretable county-level risk ranking.

The decision model integrates multiple normalized indicators, including observed obesity prevalence, key behavioral and socioeconomic risk factors, and model-derived signals from prior supervised and forecasting analyses. All criteria were standardized to ensure comparability across scales and to prevent dominance by any single variable.

Each county was assigned a composite obesity risk score representing its relative need for targeted intervention. The composite score was calculated by aggregating multiple normalized indicators, including observed obesity prevalence, key socioeconomic variables (such as poverty rate and median household income), and behavioral risk factors (notably physical inactivity). All indicators were standardized to a common scale and combined using a weighted linear aggregation, with weights informed by their relative importance observed in prior supervised learning and feature importance analyses. Higher composite scores therefore indicate counties exhibiting a convergence of elevated obesity prevalence, adverse socioeconomic conditions, and unfavorable behavioral indicators. Counties were then ranked based on this composite score to support intervention prioritization.

The resulting rankings revealed clear geographic concentration of high-risk counties, largely overlapping with regions previously identified through spatial and unsupervised analyses. This consistency across analytical stages suggests that the decision model captures structural obesity risk rather than noise-driven artifacts.

The composite risk rankings enable public health agencies to prioritize counties for intervention based on relative need

rather than uniform thresholds or statewide averages. Such prioritization supports more efficient allocation of limited resources, allowing interventions to be tailored toward counties facing the greatest combined behavioral and socioeconomic risk burden.

By translating complex, multi-dimensional data into an interpretable prioritization framework, the proposed approach bridges the gap between data mining insights and actionable public health decision-making.

Interaction effects were further examined using SHAP interaction values to assess whether key predictors jointly influence obesity risk. The interaction between physical inactivity and median household income exhibited a meaningful combined effect, indicating that the impact of inactivity on obesity prevalence varies across income levels. This result suggests that behavioral risk factors may be amplified in economically disadvantaged counties, reinforcing the need for targeted interventions.

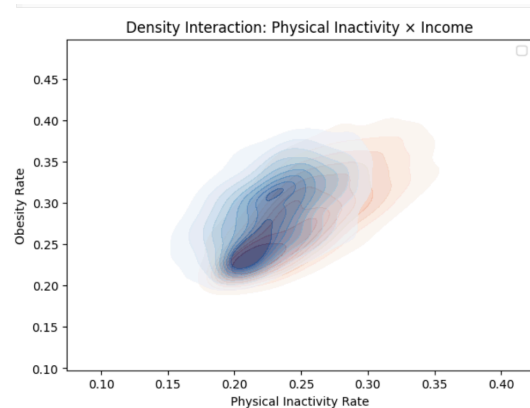


Fig.11 Interaction Between Physical Inactivity and Income in Shaping Obesity Risk

The density contours represent the joint distribution of counties across physical inactivity rates (x-axis) and obesity prevalence (y-axis), stratified by income groups. Across all income levels, obesity prevalence increases monotonically with higher physical inactivity, confirming physical inactivity as a dominant driver of obesity risk.

However, the contours reveal a clear income-dependent interaction effect. Counties with lower income levels exhibit higher obesity prevalence at comparable physical inactivity rates, as indicated by the upward shift of density regions relative to higher-income counties. In contrast, higher-income counties display lower obesity prevalence for the same levels of inactivity, suggesting a buffering effect of socioeconomic advantage.

This pattern indicates that physical inactivity alone does not fully explain obesity risk; rather, its impact is amplified under adverse socioeconomic conditions. The interaction highlights compounded vulnerability in low-income counties, where behavioral risk factors translate into disproportionately higher obesity prevalence. These findings support policy strategies that jointly address behavioral

interventions and socioeconomic constraints, rather than treating risk factors in isolation.

D. Temporal Risk Stratification via Static Clustering

While cross-sectional clustering identifies counties with similar obesity risk levels at a single point in time, it does not capture how risk evolves over time. To distinguish counties experiencing sustained deterioration from those exhibiting stable or improving trends, static clustering was applied to temporal summary features derived from multi-year data.

For each county, temporal features such as long-term slopes, mean and trend statistics were computed from historical obesity and socioeconomic indicators spanning multiple years. These features summarize whether counties exhibit increasing, stable, or decreasing obesity-related risk over time.

The clustering results revealed distinct groups of counties characterized by divergent long-term trajectories. One group exhibited persistently increasing obesity trends coupled with worsening socioeconomic conditions, while another group demonstrated relatively stable or improving trajectories. These trend-based clusters provide a complementary perspective to cross-sectional risk groupings.

Identifying counties with structurally worsening trends enables early intervention planning and informs the forecasting models presented in the subsequent section. To derive these trend-based groupings, agglomerative hierarchical clustering was applied to the temporal feature space using an average linkage criterion with a Manhattan distance metric. The number of clusters was selected as $K=2$ based on a systematic model selection pipeline that evaluated multiple clustering configurations across distance metrics and linkage strategies. Internal validation metrics, including silhouette score and Davies–Bouldin index, were used to compare candidate models and assess cluster separation and compactness. The selected configuration achieved strong performance on these metrics and produced stable, interpretable groupings.

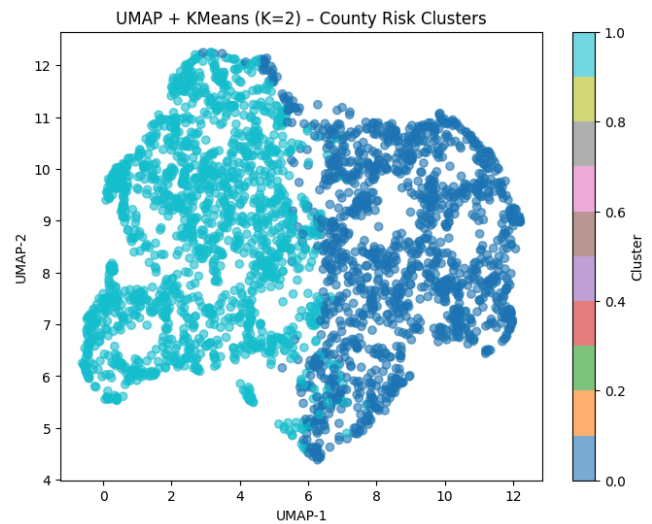


Fig. 12 UMAP projection of counties based on long-term obesity and socioeconomic trend features, colored by K-Means cluster assignment ($K=2$).

	obesity_rate	poverty_rate	physical_inactivity	median_household_income	unemployment_rate	metro_nonmetro
cluster						
0	0.307363	0.200938	0.257180	43288.166586	0.069068	Nonmetropolitan
1	0.281733	0.118487	0.216758	59576.246279	0.048375	Nonmetropolitan

Table 1. Average obesity prevalence and socioeconomic characteristics for each temporal risk cluster.

While cross-sectional clustering captures similarity at a single time point, temporal clustering distinguishes counties by direction and persistence of risk. Counties in Cluster 0 exhibit higher obesity prevalence, greater physical inactivity, and lower median income, whereas Cluster 1 represents relatively lower-risk and more stable trajectories.

E. Dynamic Joint Trend Analysis

Static trend analysis characterizes county-level obesity risk using mean obesity prevalence and mean socioeconomic and behavioral indicators. While this approach identifies structurally high- and low-risk counties, it does not capture whether obesity risk is actively changing over time or how rapidly conditions are worsening or improving.

To address this limitation, a dynamic trend analysis was conducted using temporal slopes derived from multi-year county-level data. By modeling rates of change in obesity prevalence and its associated risk factors, this approach identifies counties exhibiting accelerating risk, stabilization, or improvement, enabling early detection of emerging obesity hotspots.

Time-series patterns of obesity prevalence were jointly analyzed with corresponding trajectories of key explanatory variables, including physical inactivity and income-related indicators. This approach enables identification of counties where worsening obesity trends coincide with deteriorating behavioral or socioeconomic conditions.

The analysis revealed that counties exhibiting increasing obesity prevalence frequently experienced concurrent increases in physical inactivity and persistent socioeconomic stress. These joint trend patterns suggest reinforcing feedback mechanisms, where adverse conditions compound over time rather than acting independently.

Such dynamic co-movement highlights the importance of early intervention, as delayed responses may allow mutually reinforcing risk factors to accelerate obesity growth.

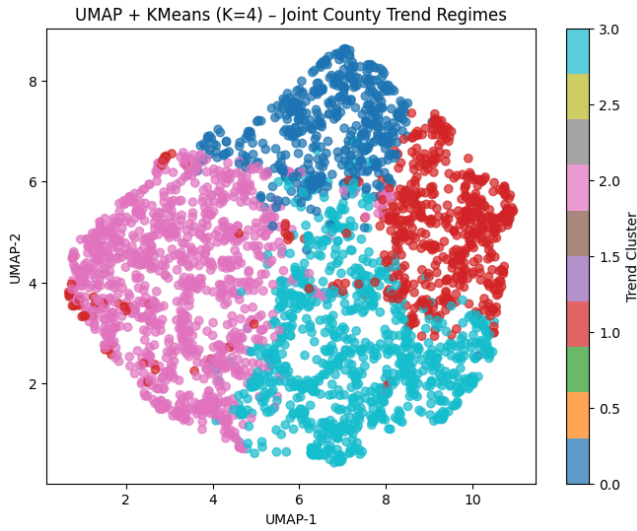


Fig. 13 UMAP visualization of county-level obesity trend regimes (2010–2023)

To capture heterogeneity in long-term obesity dynamics beyond cross-sectional risk levels, counties were clustered based on temporal summary features derived from multi-year trends, including obesity growth rates and associated socioeconomic trajectories. These features were embedded using Uniform Manifold Approximation and Projection (UMAP) to preserve non-linear structure and reduce dimensionality, and subsequently clustered using K-means to identify distinct joint trend regimes. The selection of the number of clusters was informed by a systematic clustering pipeline that evaluated multiple candidate values of K using internal validation metrics, including silhouette score, Calinski–Harabasz index, and Davies–Bouldin index. Based on this multi-metric assessment, $K=4$ was selected as providing a favorable balance between cluster separation and compactness while maintaining interpretability.

The resulting clusters reveal four qualitatively different temporal patterns. One group exhibits persistently increasing obesity prevalence accompanied by adverse socioeconomic trends, representing counties at highest long-term risk. A second group shows moderate but steady increases, indicating emerging risk. A third group demonstrates relatively stable trajectories, while a fourth group exhibits slower growth or partial stabilization over time.

These trend regimes are not purely spatially contiguous, suggesting that temporal risk evolution reflects structural socioeconomic conditions rather than geographic proximity alone. Identifying counties by trajectory regime enables early detection of worsening trends and informs the early-warning and forecasting analyses presented in subsequent sections.

F. Temporal Early-Warning Analysis

While forecasting estimates future obesity prevalence levels, it does not explicitly identify counties that are transitioning into higher-risk states. To support proactive intervention, a temporal early-warning analysis was conducted to flag counties exhibiting accelerating obesity-related risk signals before reaching critical prevalence thresholds.

Early-warning indicators were derived from temporal patterns in obesity prevalence and associated drivers, including changes in trend slopes and short-term accelerations. Counties were flagged when these indicators exceeded empirically defined thresholds, signaling elevated future risk despite moderate current prevalence.

The early-warning analysis identified a subset of counties that do not currently exhibit the highest obesity prevalence but display rapidly worsening trajectories. Many of these counties were geographically adjacent to existing high-risk regions, suggesting spatial diffusion of obesity-related risk.

From a policy perspective, early-warning counties represent high-leverage targets for intervention, as timely preventive actions may mitigate long-term health and economic costs more effectively than reactive strategies.

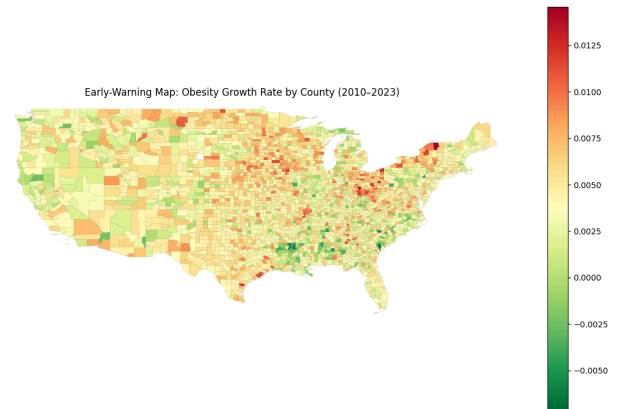


Fig. 14 Early-Warning Map of County-Level Obesity Growth Rates (2010–2023)

G. Forecasting and Long-Term Risk Projection

Long-Term Obesity Forecasts to 2030

While near-term forecasting supports validation of early-warning signals, long-term projections are necessary for strategic public health planning. To assess how current trajectories may shape future obesity risk, long-horizon

forecasts were generated to project county-level obesity prevalence through 2030.

Long-term forecasting was conducted using supervised regression models trained on historical county-level obesity prevalence and associated socioeconomic and behavioral predictors observed over multiple years. For each county, lagged obesity values and trend-based features were used to learn temporal dynamics, after which the trained model was iteratively applied to project obesity prevalence forward under a continuation of observed trends. This approach produces scenario-based projections rather than deterministic predictions, emphasizing structural persistence rather than precise point estimates.

Model performance was evaluated using out-of-sample validation, achieving an R^2 of approximately 0.66, indicating that the model explains a substantial proportion of variation in obesity prevalence, and a low root mean squared error ($RMSE = 0.00052$), reflecting stable predictive behavior at the aggregate county level. These metrics suggest that the model captures meaningful long-term signal while acknowledging inherent uncertainty in forecasting complex public health outcomes.

The 2030 projections indicate that counties already exhibiting elevated or accelerating obesity trends are likely to experience further increases over the next decade. In many cases, projected high-risk regions align with counties previously identified through spatial clustering, supervised learning, and early-warning analyses, reinforcing the persistence of structural obesity disparities over time.

From a policy perspective, long-term forecasts provide a forward-looking lens for capacity planning, enabling public health agencies to anticipate future demand and prioritize sustained intervention strategies rather than reactive responses.

V. RESULTS

This section summarizes the key findings produced by the proposed end-to-end obesity analytics framework, spanning spatial analysis, machine learning, temporal modeling, and decision support. Results are organized to reflect progressively higher levels of analytical abstraction, moving from descriptive spatial patterns to predictive and forward-looking insights.

A. Spatial Patterns of Obesity Prevalence

Spatial analysis revealed statistically significant geographic clustering of obesity prevalence across U.S. counties. Global spatial autocorrelation analysis yielded a positive Moran's I value of approximately 0.34 ($p < 0.01$), indicating that counties with similar obesity rates are geographically proximate rather than randomly distributed. Hotspot analysis further identified contiguous regions of elevated obesity prevalence, particularly in parts of the southeastern and midwestern United States, while lower-prevalence clusters were more common in coastal and highly urbanized

regions. These findings confirm that obesity prevalence exhibits strong regional dependence and motivate the need for geographically targeted intervention strategies.

B. Structural Risk Profiles from Unsupervised Learning

Unsupervised clustering of county-level features revealed distinct obesity risk profiles. High-risk clusters were characterized by elevated obesity prevalence, higher physical inactivity rates, lower median household income, and higher poverty levels, whereas low-risk clusters exhibited more favorable behavioral and socioeconomic conditions. The spatial distribution of these clusters closely aligned with previously identified obesity hotspots, indicating strong consistency between spatial dependence and latent structural risk groupings. This alignment suggests that county-level obesity disparities reflect persistent structural conditions rather than isolated or random variation.

C. Predictive Performance and Key Drivers from Supervised Learning

Supervised learning models were applied to quantify the predictive relationship between obesity prevalence and associated socioeconomic and behavioral risk factors. The best-performing models explained approximately 40% of the variance in county-level obesity prevalence on held-out data, demonstrating meaningful predictive capability given the multifactorial nature of obesity. Feature importance and SHAP analyses consistently identified physical inactivity as the most influential predictor, followed by socioeconomic indicators such as median household income and poverty rate. Interaction analysis further revealed that the effect of physical inactivity on obesity prevalence is amplified in lower-income counties, highlighting compounding risk mechanisms across behavioral and economic dimensions.

Predicted Obesity Risk Priority (Composite Risk Score)

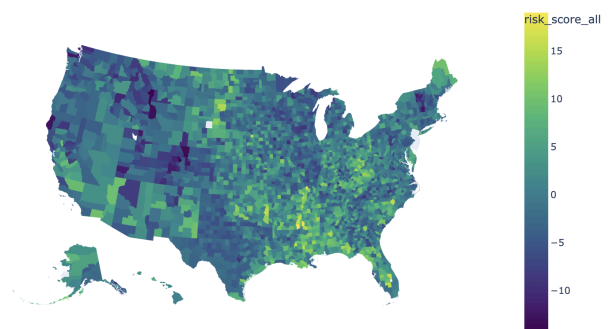


Fig. 15 County-level composite obesity risk priority score derived from multi-criteria decision modeling. Higher values indicate greater intervention priority based on current prevalence, socioeconomic vulnerability, behavioral risk factors, and temporal trends.

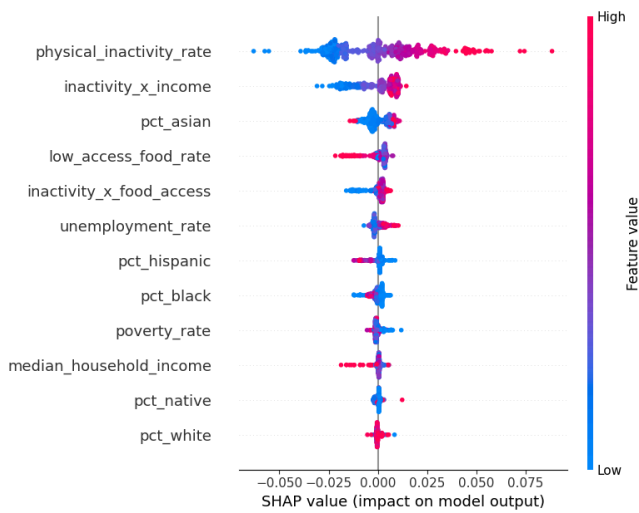


Fig. 16 presents a SHAP interaction analysis illustrating how physical inactivity and median household income jointly influence county-level obesity prevalence predictions. SHAP values quantify each feature’s contribution to the model’s output relative to a baseline prediction, allowing both main effects and interaction effects to be examined.

The visualization indicates that higher physical inactivity rates consistently contribute positively to predicted obesity prevalence. However, the magnitude of this effect varies with income level. Counties with lower median household income exhibit larger positive SHAP values for physical inactivity, indicating that inactivity contributes more strongly to obesity risk in economically disadvantaged settings. In contrast, for higher-income counties, the marginal impact of physical inactivity on predicted obesity prevalence is attenuated.

This interaction suggests that socioeconomic context moderates behavioral risk factors: identical levels of physical inactivity lead to higher obesity risk in lower-income counties than in higher-income counties. The result highlights compounding risk mechanisms and supports the need for targeted interventions that jointly address behavioral and socioeconomic determinants rather than treating them independently.

To further quantify the aggregate impact of the identified interaction, a baseline prediction was computed by averaging model outputs across all counties using the trained supervised model. This baseline represents the expected county-level obesity prevalence under observed conditions and yielded a mean predicted value of approximately 0.288. A counterfactual simulation was then performed in which key interaction conditions were altered, and the resulting mean prediction was compared against the baseline.

The difference between the baseline mean prediction and the simulated mean prediction corresponds to an average impact

of approximately 0.016 in obesity prevalence. Although modest in absolute magnitude, this shift represents a systematic population-level effect attributable to the interaction between physical inactivity and socioeconomic context. The result demonstrates that interaction effects identified through SHAP analysis translate into meaningful aggregate changes in predicted obesity prevalence, reinforcing the importance of considering compounding behavioral and socioeconomic factors in policy-oriented obesity risk assessment.

D. Temporal Risk Stratification and Early-Warning Detection

Temporal analysis extended beyond cross-sectional patterns to identify counties exhibiting worsening obesity trajectories over time. Static clustering of long-term trend features differentiated counties with persistently increasing obesity prevalence from those with stable or improving trends. Building on this, a temporal early-warning system identified counties that do not yet exhibit the highest obesity prevalence but show accelerating risk signals. Many early-warning counties were geographically adjacent to existing high-risk regions, suggesting potential spatial diffusion of obesity-related risk over time.

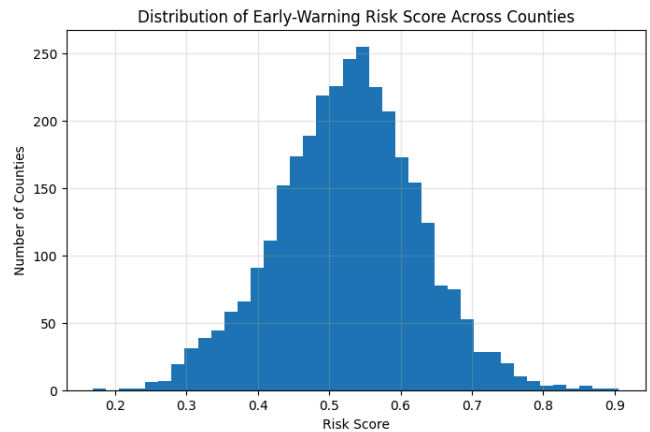


Fig. 17 presents distribution of early warning risk score across counties

To select the number of clusters and linkage strategy for agglomerative clustering, a grid search was conducted across multiple linkage criteria and cluster counts. Model quality was evaluated using silhouette score, Davies–Bouldin index, and Calinski–Harabasz score. The configuration with two clusters and average linkage achieved the best overall performance (silhouette ≈ 0.64 , Davies–Bouldin ≈ 0.57), and was therefore selected for subsequent analysis.

Static Clustering

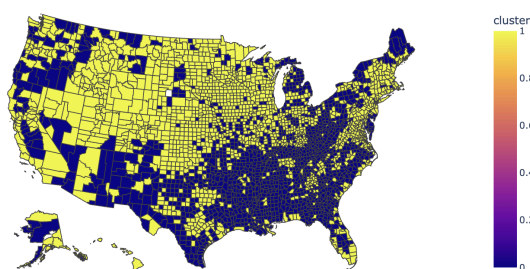


Fig 18 presents static clustering results across the United States. Yellow showing low obesity and blue as high.

The resulting cluster solution reveals a clear bifurcation between counties experiencing persistently worsening obesity trajectories and those exhibiting relatively stable or slower growth patterns. This binary stratification provides a robust foundation for early-warning identification, distinguishing counties requiring immediate intervention from those with lower short-term risk.

E. Forecasting and Long-Term Risk Projection

Forecasting analysis provided both near-term and long-term perspectives on county-level obesity risk. Short-term forecasts for early-warning counties indicated continued increases in obesity prevalence, validating the early-warning signals. Long-term projections through 2030 suggest that counties already exhibiting elevated or accelerating risk are likely to experience further increases if current trends persist. Notably, projected high-risk regions aligned closely with counties previously identified through spatial analysis, clustering, and supervised modeling, reinforcing the internal consistency and robustness of the proposed framework.

To assess whether obesity growth trajectories shifted following the COVID-19 period, county-level obesity growth slopes were computed separately for pre-2019 and post-2020 intervals. Figure 17 compares these slopes across counties. Most counties cluster near the origin, indicating broadly similar growth rates before and after COVID-19. However, a subset of counties exhibit positive post-COVID growth despite relatively stable pre-COVID trends, suggesting localized accelerations in obesity prevalence. These findings motivate the use of trend-based clustering and early-warning indicators rather than assuming a uniform nationwide structural break.

Pre vs Post COVID Obesity Growth by County

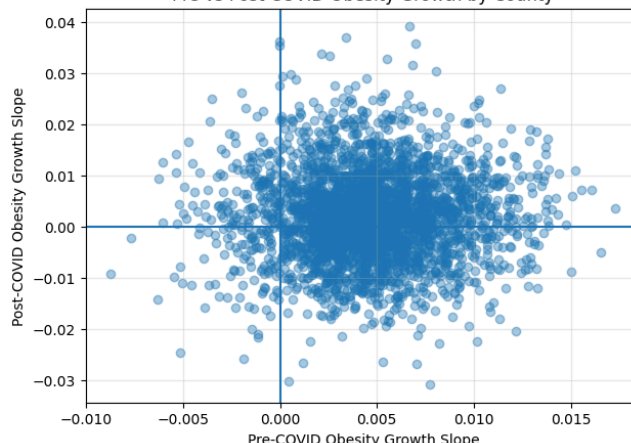


Fig.19 Pre and Post-COVID Obesity Growth Dynamics Across U.S. Counties

Early-Warning Risk Tiers for Obesity Growth by 2030

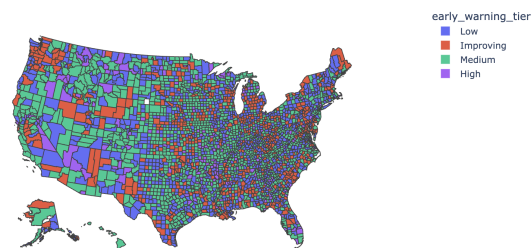


Fig 20 County-Level Early-Warning Risk Tiers for Obesity Growth by 2030

Building on the continuous early-warning risk scores, counties were categorized into discrete early-warning tiers to facilitate interpretability and policy relevance. Figure 19 presents county-level early-warning risk tiers for projected obesity growth through 2030. Counties were classified into four categories - Low, Improving, Medium, and High based on the magnitude and consistency of temporal risk signals and forecasted trends.

High-risk counties exhibit sustained and accelerating obesity growth trajectories, often adjacent to existing high-prevalence regions, indicating potential spatial diffusion of risk. Medium-risk counties display moderate but persistent upward trends, while improving counties show stabilizing or declining trajectories despite historical risk. Low-risk counties demonstrate consistently stable obesity patterns over time.

This tiered representation translates complex temporal analytics into an interpretable early-warning system, enabling proactive identification of counties where intervention may prevent future escalation rather than reacting to already elevated prevalence.

F. Decision-Support Outcomes

The integration of analytical outputs into a multi-criteria decision-making (MCDM) framework produced a composite obesity risk ranking for counties nationwide. Counties ranked highest consistently exhibited a combination of elevated current prevalence, adverse socioeconomic conditions, and unfavorable temporal trajectories. This prioritization framework translates complex, multi-dimensional analytical results into an interpretable and actionable decision-support tool, enabling public health agencies to allocate intervention resources more effectively and proactively.

VI. DISCUSSION

This study demonstrates that county-level obesity in the United States is driven by interacting behavioral, socioeconomic, and spatial factors rather than isolated determinants. The alignment of findings across exploratory analysis, spatial statistics, unsupervised clustering, supervised learning, and temporal forecasting indicates that observed obesity disparities reflect persistent structural conditions. Counties identified as high-risk through one analytical method were consistently reinforced by others, supporting the robustness of the proposed framework.

A central finding is the dominant role of physical inactivity in explaining county-level obesity prevalence, particularly when interacting with socioeconomic disadvantage. Supervised learning and interaction analysis revealed that the impact of physical inactivity is amplified in lower-income counties, suggesting compounding risk mechanisms. These results highlight that behavioral interventions alone may be insufficient without addressing underlying economic and environmental constraints.

The inclusion of temporal trend analysis and early-warning detection extends obesity surveillance beyond descriptive reporting. Counties not currently exhibiting the highest obesity prevalence but showing accelerating risk trajectories were successfully identified, enabling proactive intervention planning. Long-term forecasting further indicates that, in the absence of targeted intervention, existing regional disparities are likely to persist or intensify. Collectively, these findings underscore the value of integrating spatial, predictive, and temporal analytics to inform geographically targeted and forward-looking public health strategies.

VII. LIMITATIONS AND FUTURE WORK

Several limitations should be considered when interpreting the results of this study. First, the analysis relies on county-level aggregate data, which limits insight into individual-level behaviors and introduces the possibility of ecological fallacy. Second, although a broad range of socioeconomic and behavioral variables were included, dietary intake, physical infrastructure, and built-environment characteristics were not consistently available at the national scale and therefore could not be incorporated. Third, all relationships identified in this study are observational, and causal inference cannot be established from the current modeling approach.

Additionally, forecasting models assume continuity of historical trends and do not account for future policy interventions, economic shocks, or public health initiatives that may alter obesity trajectories. While forecasting performance is stronger for long-term trend estimation than cross-sectional prediction, uncertainty remains in projected outcomes.

Future work may extend this framework by incorporating individual-level or sub-county data where available, enabling stronger causal modeling and more granular risk assessment. Integrating built-environment features such as walkability, food accessibility, and transit infrastructure may further improve explanatory power. Finally, applying causal inference or policy evaluation models could allow estimation of intervention impacts, transforming the framework from risk identification toward outcome-driven policy assessment.

VIII. CONCLUSION

This paper presented a comprehensive data-driven framework for county-level obesity risk identification, forecasting, and public health decision support in the United States. By integrating exploratory data analysis, spatial statistics, machine learning, temporal modeling, and multi-criteria decision analysis, the proposed approach advances obesity surveillance beyond descriptive mapping toward actionable, policy-relevant insights. The results reveal persistent geographic disparities, identify emerging high-risk regions, and project future obesity trends under current conditions. Overall, this framework provides a scalable and interpretable foundation for targeted, cost-effective, and evidence-based public health intervention planning.

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